

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (ME- CAD/CAM) Examination

Dec. 2013- Jan. 2014

ME702.01 Advances in Manufacturing Technology

Date: 03.01.2014, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer following multiple choice type questions. There could be more than one correct answer. Write all correct choices in answer sheet. [10]

- (i) Volumetric contraction between pouring and liquidus temperature is mainly compensated by:
- | | |
|------------------------|-----------------------|
| (p) Pattern allowances | (q) Gating system |
| (r) Feeding system | (s) None of the above |
- (ii) Which of the following is most ideal shape of the feeder:
- | | |
|---------------|------------------------|
| (p) Cuboid | (q) Cylindrical |
| (r) Spherical | (s) Hollow cylindrical |
- (iii) Favorable conditions for mold erosion are:
- | | |
|--------------------------|-------------------------------|
| (p) High impact velocity | (q) High density molten metal |
| (r) High mold strength | (s) High sprue length |
- (iv) During casting solidification, the heat transfer coefficient at metal-mold interface is the lowest:
- | | |
|--|----------------------------------|
| (p) At the beginning of solidification | (q) At the end of solidification |
| (r) At the end of 50% solidification | (s) None of the above |
- (v) Stresses on the mold due to core weight and buoyancy force will depends on:
- | | |
|--------------------------------------|---------------------------------------|
| (p) Core surface area | (q) Core print surface area |
| (r) Core and core print surface area | (s) Core print projected surface area |
- (vi) The advantages of investment casting include:
- | | |
|---|--|
| (p) Useful to very small to very large castings | (q) Intricate and thin features possible |
| (r) High surface finish and tolerances | (s) All of the above |
- (vii) A fusion welding process using chemical energy source is:
- | | |
|---------------------|-----------------------|
| (p) Gas welding | (q) Induction welding |
| (r) Thermit welding | (s) Arc welding |
- (viii) During welding process, metal shielding is provided to prevent:
- | | |
|---------------|----------------------|
| (p) Oxidation | (q) Weld inclusions |
| (r) Arc blow | (s) All of the above |
- (ix) Which of the followings is(are) not solid state welding processes
- | | |
|------------------------|----------------------|
| (p) Resistance welding | (q) Gas welding |
| (r) Ultrasonic welding | (s) Pressure Welding |
- (x) In resistance welding, the resistance should be highest:
- | | |
|----------------------|--|
| (p) Inside electrode | (q) At interface between workpiece and electrode |
| (r) Inside workpiece | (s) At interface between two workpieces |

- Q - 2 The voltage-length characteristic of a direct current arc is given by, $V = 10 + 20l$ volts, [07]
where l is length of arc in cm. The power source characteristic can be approximated by a straight line with an open circuit voltage 50 V and a short circuit current 600 amp. Determine the optimal arc length and power.

OR

- Q - 2 For an aluminum alloy cuboid casting, having 20 X 30 X 40 cm dimensions, poured into a silica sand mold, compute the solidification time if molten metal is poured at 100 °C superheat temperature. Take $k_{mold} = 0.7 \text{ W/m-K}$, $\rho_{mold} = 1600 \text{ Kg/m}^3$, $C_{mold} = 1000 \text{ J/Kg-K}$, $\rho_{metal} = 2600 \text{ kg/m}^3$, $C_{metal} = 800 \text{ J/kg-K}$, $\Delta H_{metal} = 400 \text{ KJ/Kg}$, $T_{melting} = 600^\circ\text{C}$, and ambient temperature = 30 °C. [07]

- Q - 3 Attempt any three: [18]
- Describe the concept of Virtual Manufacturing with an illustration.
 - Discuss and elaborate the application areas of thin films.
 - Discuss the influence of pouring at under superheat and over superheat temperature on castings.
 - Briefly discuss various sources of heat that can be used to perform welding operation.

SECTION – II

- Q - 4 Derive the equation for MRR in abrasive jet machining with a neat sketch. [08]
- Q - 5 A Company manufacturing a bearing of standard 6003 .After making a little change in production cycle, they received number of complaints from customers. Company wants to change the inspection method to make good relation with customers. The different part batches inspected by the three different quality inspection methods are given in table below. Comment if hypothesis methods are best indicator of the quality or not? ($\alpha=0.05$) [07]

Method	1	2	3	4	5
X1	2	6	8	5	4
X2	3	8	7	3	1
X3	5	4	3	2	5

OR

- Q - 5 (a) Explain randomization with respect to design of experiments by suitable example? [04]
- Q - 5 (b) Explain the concept of blocking in full factorial design. [03]
- Q - 6 Attempt any four: [20]
- Explain the use of stratified wire in wire-EDM
 - Explain the Laser beam generation process
 - Discuss the working principle of Plasma arc machining with a neat sketch. State its advantages and limitations.
 - Describe the working principle of ultrasonic machining process and discuss the types of ultrasonic transducers.
 - Discuss the steps involved in planning, executing, analyzing and reporting the experiments.
